Arc Length and Curvature

section 12.5

10-14-26

$$S = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}$$

$$S = \int_a \sqrt{(r'(t))^2}$$

HW 12.5 10/14

9) $r(t) = -ti + 4tj + 3tk$

$v(t) = \langle -1, 4, 3 \rangle$

$\|v(t)\| = \sqrt{26}$

$$\int_0^1 \sqrt{26}\, dt = \sqrt{26}t \Big]_0^1 = \sqrt{26}$$

10) $r(t) = i + t^2 j + t^3 k$

$v(t) = \langle 0, 2t, 3t^2 \rangle$

$\|v(t)\| = \sqrt{4t^2 + 9t^4} = t\sqrt{4+9t^2}$

$$\frac{1}{18}\int_0^2 18t\sqrt{4+9t^2}\, dt = \frac{1}{18}\cdot\frac{2}{3}(4+9t^2)^{3/2}\Big]_0^2$$

$$= \frac{1}{27}\left[40^{3/2} - 4^{3/2}\right]$$

$$= \frac{8}{27}\left[10^{3/2} - 1\right]$$

17) $= \langle 2\cos t, 2\sin t, t \rangle$

a) $r'(t) = \langle -2\sin t, 2\cos t, 1 \rangle$

b) $r(s) = \langle 2\cos(\frac{s}{\sqrt{5}}), 2\sin(\frac{s}{\sqrt{5}}), \frac{s}{\sqrt{5}} \rangle$

$r'(s) = \langle -\frac{2}{\sqrt{5}}\sin(\frac{s}{\sqrt{5}}), \frac{2}{\sqrt{5}}\cos(\frac{s}{\sqrt{5}}), \frac{1}{\sqrt{5}} \rangle$

$r''(s) = \langle -\frac{2}{5}\cos(\frac{s}{\sqrt{5}}), -\frac{2}{5}\sin(\frac{s}{\sqrt{5}}), 0 \rangle$

$\|r'(s)\| = \frac{a}{a^2+b^2}$

B) Arc Length Function

$$\frac{ds}{dt} = \|r'(t)\|$$

$$\int ds = \int \|r'(t)\|\, dt \qquad S = \int_a^t \|r'(t)\|\, dt$$

---

ex) $r(t) = \langle 3-3t, t, 0 \rangle$

$v(t) = \langle -3, 1, 0 \rangle$

$\|v(t)\| = \sqrt{10}$

$$S = \int_0^t \sqrt{10}\, dt \qquad S = t\sqrt{10} \qquad t = \frac{s}{\sqrt{10}}$$

$r(s) = \langle 3 - 3\cdot\frac{s}{\sqrt{10}}, \frac{s}{\sqrt{10}}, 0 \rangle$

C) Curvature

$$K = \|T'(s)\| = \|r''(s)\|$$

HW 12.5 10/15

26) $K = \frac{\|r' \times r''\|}{\|r'\|^3}$ $\qquad r(t) = ti + \frac{1}{9}t^3 j \qquad t=2$

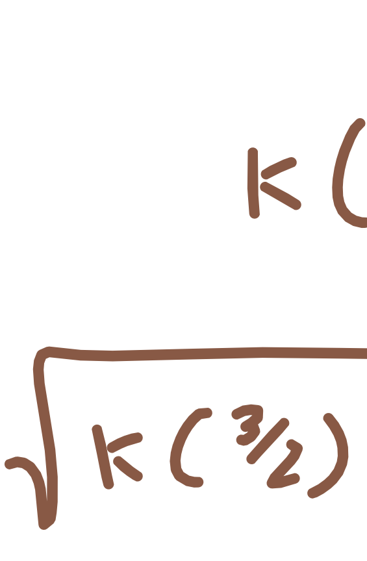
$r = \langle t, \frac{1}{9}t^3, 0 \rangle \qquad K = \frac{\|v \times a\|}{\|v\|^3}$

$v = r' = \langle 1, \frac{1}{3}t^2, 0 \rangle$

$a = r'' = \langle 0, \frac{2}{3}t, 0 \rangle$

$$K = \frac{\|\langle 1, 4/3, 0 \rangle \times \langle 0, 4/3, 0 \rangle\|}{(5/3)}$$

$$K = \frac{4/3}{5/3} = 4/5$$



$$S = \frac{c}{\sqrt{K}}$$

$y = \frac{1}{3}x^3 \qquad y' = x^2 \qquad y'' = 2x$

68) $K = \frac{|y''|}{[1+(y')^2]^{3/2}} \qquad \frac{2}{2^{3/2}} = \frac{1}{\sqrt{2}}$

$$K(\tfrac{3}{2}) = \frac{3}{(1+\frac{3^4}{2^4})^{3/2}}$$

$$\sqrt{K(3/2)} = \left(\frac{3}{(1+\frac{3^4}{2^4})^{3/2}}\right)^{1/2} \qquad S = \frac{c}{\sqrt{K}}$$

$$S = \frac{30}{2^{1/2}}\;\frac{(2^4+3^4)^{3/4}}{3^{1/2}2^3}$$